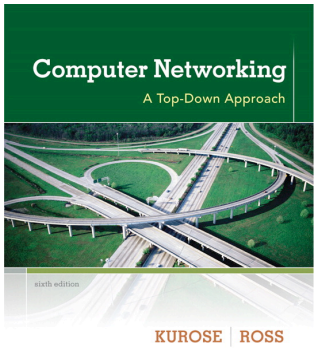
Wireshark Lab: TCP
SOLUTION



Supplement to *Computer Networking: A Top-Down Approach, 6th ed.,* J.F. Kurose and K.W. Ross

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The answers below are based on the trace file *tcp-ethereal-trace-1* in in http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip

TCP Basics

Answer the following questions for the TCP segments:

1. *What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?*
2. *What is the IP address and port number used by gaia.cs.umass.edu to receive the file.*

Solution: Client computer (source)
IP address: `192.168.1.102`
TCP port number: `1161`

Destination computer: gaia.cs.umass.edu
IP address: `128.119.245.12`
TCP port number: `80`

3. If you did this problem on your own computer, you'll have your own solution

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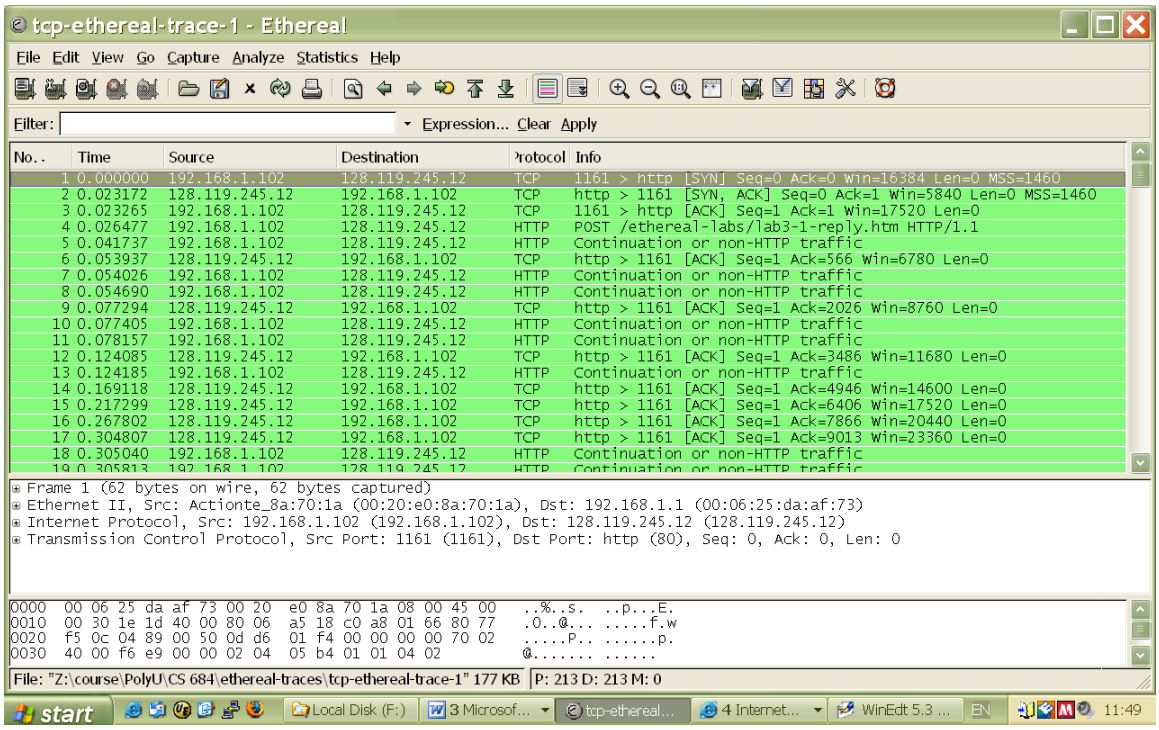

Figure 1: IP addresses and TCP port numbers of the client computer (source) and gaia.cs.umass.edu

4. *What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?*

Solution: Sequence number of the TCP SYN segment is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu. The value is 0 in this trace.

The SYN flag is set to 1 and it indicates that this segment is a SYN segment.

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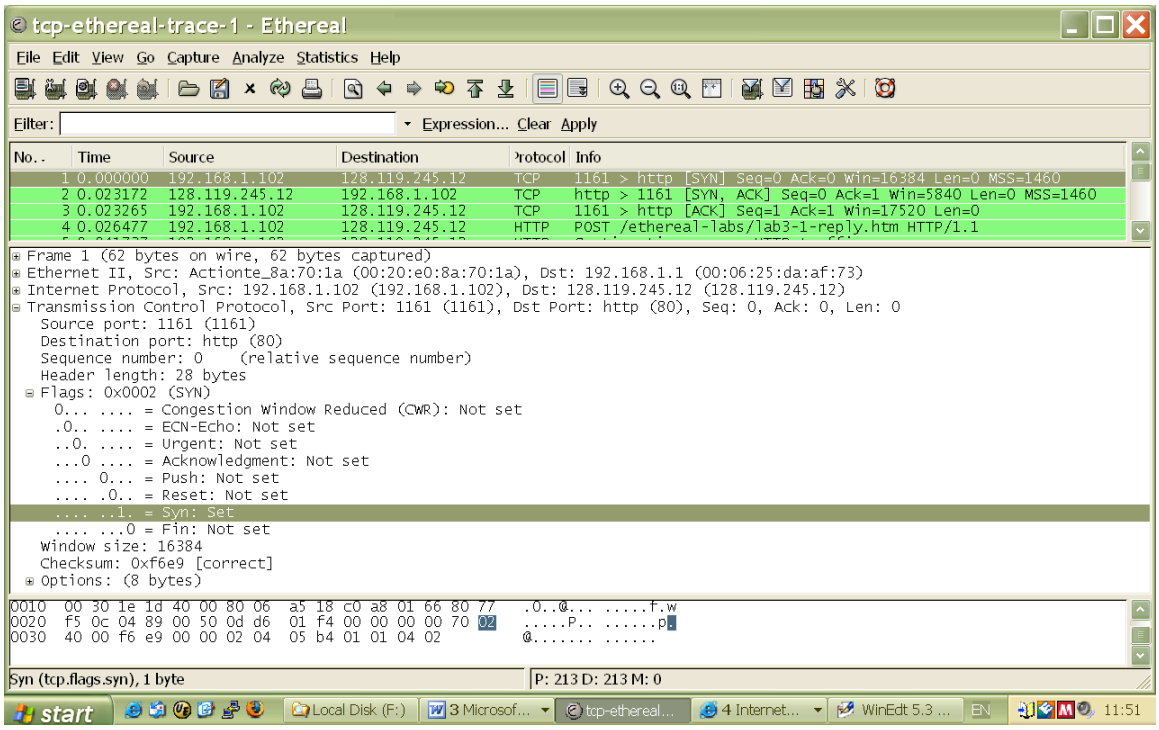

Figure 2: Sequence number of the TCP SYN segment

5. *What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the ACKnowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?*

Solution: Sequence number of the SYNACK segment from gaia.cs.umass.edu to the client computer in reply to the SYN has the value of 0 in this trace.

The value of the ACKnowledgement field in the SYNACK segment is 1. The value of the ACKnowledgement field in the SYNACK segment is determined by gaia.cs.umass.edu by adding 1 to the initial sequence number of SYN segment from the client computer (i.e. the sequence number of the SYN segment initiated by the client computer is 0.).

The SYN flag and Acknowledgement flag in the segment are set to 1 and they indicate that this segment is a SYNACK segment.

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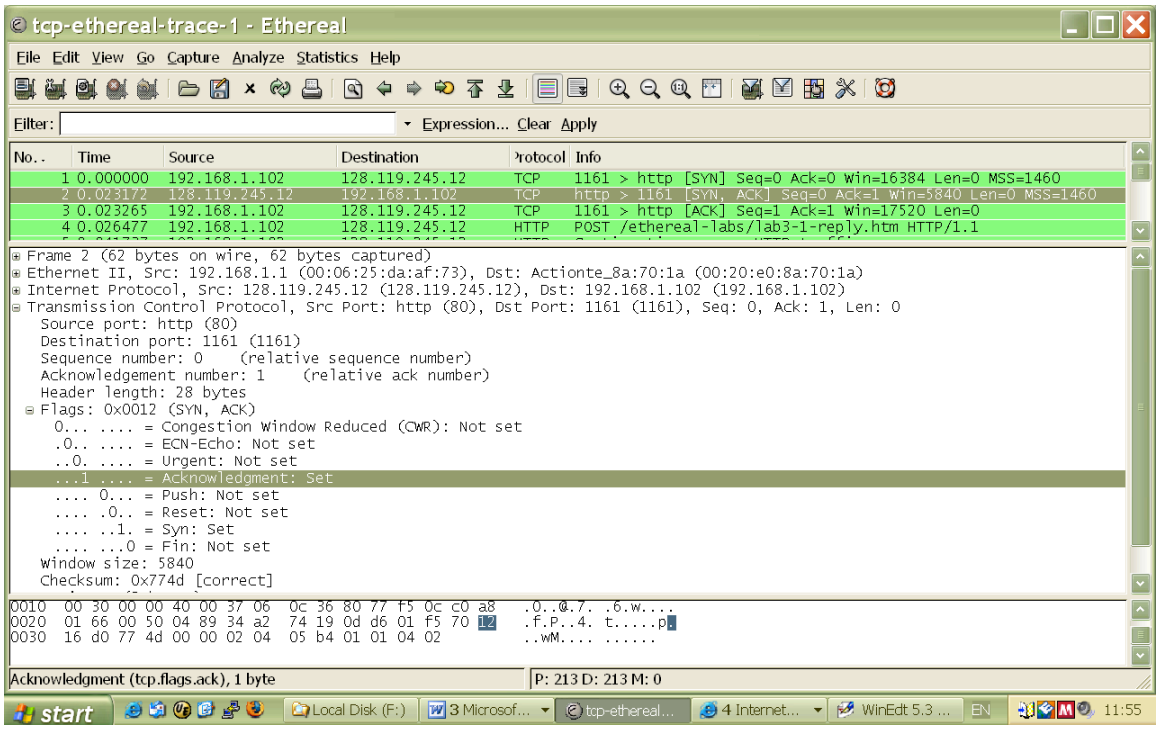

Figure 3: Sequence number and Acknowledgement number of the SYNACK segment

6. *What is the sequence number of the TCP segment containing the HTTP POST command? Note that in order to find the POST command, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with a "POST" within its DATA field.*

Solution: No. 4 segment is the TCP segment containing the HTTP POST command. The sequence number of this segment has the value of 1.

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